Paper IV – Recent Advances (MCh Urology, Summer 2017)

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IV-1. Describe in brief intravesical therapy for transitional cell carcinoma of Urinary bladder.

Type: LAQ (25 marks)

Introduction

- NMIBC (non-muscle-invasive bladder cancer: Ta, T1, CIS [carcinoma in situ]) is 70–80% of bladder cancers. "Urothelial carcinoma" now replaces "transitional cell carcinoma".
- Two problems: **recurrence** (40–60%) and **progression** (about 6% in low-grade Ta to 17% in high-grade T1). Intravesical therapy follows complete TURBT (transurethral resection of bladder tumour).
- **Rule:** chemotherapy reduces recurrence only. BCG (bacille Calmette-Guérin) is the only agent that also reduces progression.

Risk-adapted use (AUA [American Urological Association]/SUO)

Risk	Criteria	Therapy
Low	LG (low-grade) solitary Ta ≤3 cm; PUNLMP	Single post-op instillation only
Intermediate	LG Ta recurrent, >3 cm or multifocal; LG T1; HG (high-grade) Ta ≤3 cm	6-week chemotherapy or BCG; maintenance 1 year
High	HG T1; HG Ta large, multifocal or recurrent; any CIS; LVI (lymphovascular invasion); variant histology	BCG induction + maintenance 3 years

- EAU (European Association of Urology) adds a "very high" group; EORTC (European Organisation for Research and Treatment of Cancer) and CUETO risk tables.

1. Single immediate instillation

- **Rationale:** kills floating cells before they implant. So it must be early: **within 6 hours** (no benefit at 24 hours).
- Agents: MMC (mitomycin C) 40 mg, epirubicin 50 mg, **gemcitabine 2 g** (standard in USA). **SWOG S0337:** 4-year recurrence 35% vs 47% with saline.
- **Withhold** after perforation or extensive resection. **Never give BCG** now (intravasation, sepsis).

2. Adjuvant chemotherapy

- **MMC** (alkylating), 40 mg in 20 mL weekly ×6. Optimise it (Au, 2001): empty bladder, fluid restriction, sodium bicarbonate, concentrated dose. Skin rash.
- **Gemcitabine** (nucleoside analogue): minimal absorption, well tolerated. **Anthracyclines** (epirubicin, doxorubicin) cause chemical cystitis.
- Intermediate risk: 6-week induction, then monthly maintenance for 1 year.
- **Gem/Doce:** gemcitabine 1 g, drain, then docetaxel 37.5 mg; weekly ×6, then monthly for 2 years. Used in BCG-unresponsive disease and during BCG shortage.

3. BCG immunotherapy

- **Mechanism:** BCG binds **fibronectin**. Cytokines follow (IFN-γ [interferon-gamma], IL [interleukin]-2, IL-12, TNF-α [tumour necrosis factor-alpha]). A **Th1 (T-helper 1)** response gives cell-mediated tumour kill; TRAIL release is important.
- **Technique:** 1 vial in 50 mL saline. Start **2–4 weeks** after TURBT (re-epithelialisation). Urinalysis first; retain 1–2 hours.
- **SWOG schedule (Lamm, 2000):** 6 weekly induction doses, then 3 weekly doses at 3 and 6 months, then every 6 months to 3 years. Median recurrence-free survival 76.8 vs 35.7 months.
- **EORTC 30962:** full dose; 1 year for intermediate, 3 years for high risk. **NIMBUS:** reduced frequency was inferior.
- CIS complete response up to 84%; BCG beats MMC only with maintenance.
- **Contraindications:** immediately after TURBT, gross haematuria, traumatic catheterisation (all risk intravasation); previous BCG sepsis; total incontinence; immunosuppression (now relative). Relative: active UTI (urinary tract infection), liver disease, previous TB (tuberculosis).

BCG toxicity

Grade	Features	Management
1	Cystitis, fever <38.5 °C, <48 h	Anticholinergics, analgesics, NSAIDs (non-steroidal anti-inflammatory drugs)
2	Severe or >48 h	Culture; isoniazid + rifampicin; resume at ½–⅓ dose
3	Sepsis, organ involvement	Stop BCG; isoniazid + rifampicin + ethambutol; steroids only with cover

- BCG is resistant to **pyrazinamide**. BCG sepsis 0.5%.

BCG failure

- **Adequate BCG:** ≥5 of 6 induction + ≥2 of 3 maintenance doses.
- **BCG-unresponsive:** persistent or recurrent CIS or HG Ta/T1 despite adequate BCG, or HG relapse within 6 months (Ta/T1) or 12 months (CIS) of the last dose.
- **Also know:** refractory, intolerant, exposed, delayed relapse.

BCG-unresponsive options

- **Timely radical cystectomy** is the standard (EAU, AUA).
- If unfit or refusing:
  - **Pembrolizumab** (KEYNOTE-057): IV every 3 weeks; CIS complete response 41%.
  - **Nadofaragene firadenovec:** adenovirus with interferon-α2b gene, every 3 months; CIS complete response 53%.
  - **N-803:** IL-15 superagonist with BCG; CIS complete response 71%.
  - Gem/Doce; valrubicin; **TAR-200** (gemcitabine-releasing device); cretostimogene (trials).

Device-assisted therapy

- **EMDA (electromotive drug administration):** current increases MMC tissue delivery 4–7-fold. Sequential BCG + EMDA-MMC improved outcomes in T1.
- **Chemohyperthermia:** RITE (radiofrequency, 42 °C); HIVEC (heated recirculating MMC).
- **Also know:** UGN-102 MMC gel; interferon with low-dose BCG.

Diagram to draw: BCG mechanism (fibronectin → cytokines → Th1 → kill) with SWOG schedule timeline.

> **Key points to remember:**
> - Single instillation within 6 h; never BCG immediately (SWOG S0337 gemcitabine).
> - Only maintenance BCG reduces progression; SWOG 3-year schedule.
> - BCG-unresponsive: cystectomy; pembrolizumab, nadofaragene, N-803.

IV-2. Describe applications of various LASERs in urology.

Type: LAQ (25 marks)

Principle

- LASER (light amplification by stimulated emission of radiation): a "pumped" gain medium releases identical photons. A mirror resonator amplifies them, and a partial mirror lets the beam out.
- Light is **monochromatic, coherent and collimated**, so it passes down thin fibres.
- **Chromophore** decides the effect. Water absorbs holmium and thulium light. Haemoglobin absorbs KTP (potassium titanyl phosphate) light.
- Absorption sets **penetration**. Shallow penetration vaporises and cuts. Deep penetration coagulates and later sloughs.
- Coagulation occurs at about 60 °C; vaporisation at ≥100 °C.
- **Pulse duration** decides the stone mechanism. Short pulses (<10 µs) act photomechanically (shock waves). Long pulses (>250 µs: Ho:YAG [holmium:yttrium-aluminium-garnet], TFL [thulium fibre laser]) act **photothermally**.

Lasers used

Laser	Wavelength (nm)	Chromophore	Penetration	Main uses
Ho:YAG	2120 (Ch 19 quotes 2100)	Water	0.3–0.4 mm	Stones, HoLEP, incisions, tumours
TFL	1940	Water (peak)	<0.1 mm	Dusting, enucleation
Tm:YAG	~2010	Water	0.2 mm	Vaporesection, enucleation
KTP/LBO	532	Haemoglobin	0.8 mm	PVP
Diode	940–1470	Water + haemoglobin	0.5–5 mm	Prostate vaporisation
Nd:YAG	1064	Neither strongly	Up to 10 mm	Coagulation of tumours
CO_2	10,600	Water	~0.1 mm	Skin and penile lesions

1. Stones

- **Ho:YAG** is the gold standard. It breaks all stone types. Used in URS (ureteroscopy), RIRS (retrograde intrarenal surgery), PCNL (percutaneous nephrolithotomy) and for bladder stones.
- **Moses effect:** the start of each pulse makes a vapour channel; the rest crosses it to the stone. So keep the fibre close and perpendicular.
- Fragmenting: 0.6–1.0 J × 6–10 Hz. Dusting: 0.2–0.3 J × 40–80 Hz.
- **Retropulsion** rises with pulse energy and falls with longer pulses or pulse modulation.
- **TFL:** higher water absorption and low peak power give finer dust and less retropulsion. It beat low-power holmium in RCTs (randomised controlled trials) and equalled 120 W pulse-modulated holmium.
- **Heat** depends on power, not laser type. Keep ≤10 W in the ureter; use good flow and an access sheath.

2. BPH (benign prostatic hyperplasia)

- **HoLEP (holmium laser enucleation of prostate):**
  - Endoscopic equivalent of open prostatectomy, **size-independent**. Enucleate at about 2 J × 40–50 Hz, then morcellate.

    - Only endoscopic procedure superior to TURP (transurethral resection of prostate). Less bleeding, shorter catheter time.
- **PVP (photoselective vaporisation, KTP/LBO 120–180 W):**
    - Haemoglobin absorption gives bloodless vaporisation, so anticoagulants can continue.
    - **GOLIATH RCT:** non-inferior to TURP. Avoid >80 cc (retreatment).
- **Thulium:** vaporisation, vaporesection, ThuLEP (equivalent to HoLEP). Diode vaporisation also used.
- Nd:YAG coagulation was abandoned (1 cm penetration, sloughing).
- AUA (American Urological Association): HoLEP, ThuLEP or PVP for **high bleeding risk**.

3. Incisions

- Laser urethrotomy has no advantage over cold knife.
- Holmium incision of bladder neck contracture, ureteric stricture, endopyelotomy, ureterocele and calyceal diverticulum neck.

4. Urothelial tumours

- **NMIBC (non-muscle-invasive bladder cancer):** laser **en bloc resection** (Ho, Tm, KTP) gives detrusor in the specimen and no obturator jerk. Office ablation of small low-grade recurrences.
- **UTUC (upper tract urothelial carcinoma):** biopsy, then ablate by ureteroscopy:
    - Ho:YAG: <0.5 mm, haemostatic.
    - Nd:YAG: 5–6 mm coagulation, so stricture risk.
    - Thulium: <0.2 mm.
- **PDT (photodynamic therapy):** a photosensitiser plus laser light makes singlet oxygen.

5. Penile and genital lesions

- PeIN (penile intraepithelial neoplasia) and small Ta/T1: **Nd:YAG preferred**, with base biopsies.
- CO_2 is too superficial for T1 (recurrence up to 50%). CO_2 or Nd:YAG for genital warts.

Safety

- Wavelength-specific eyewear for all, including the patient. Door signage, covered windows, trained staff and a laser safety officer.
- Check fibres; never fire inside the scope. Guard against fire and thermal injury.

Recent advances

- Pulse modulation (Moses) and high-frequency holmium. TFL and thin fibres. Suction sheaths. AI (artificial intelligence) settings.

Diagram to draw: water and haemoglobin absorption curves marking 532, 1064, 1940 and 2120 nm.

> **Key points to remember:**
> - Chromophore sets penetration: TFL shallowest, Ho 0.4 mm, Nd:YAG deepest.
> - Holmium lithotripsy is photothermal (Moses); TFL gives less retropulsion.
> - HoLEP size-independent, superior to TURP; PVP for anticoagulated men (GOLIATH).
> - Nd:YAG for PeIN; eyewear for everyone.

IV-3a. Prostate cancer prevention trials

Type: SAQ (10 marks)

Rationale

- Prostate cancer is common, slow and androgen-dependent. Men with 5α-reductase deficiency have rudimentary prostates. This supported **5ARI (5α-reductase inhibitor)** trials.

Landmark trials

	PCPT (2003)	REDUCE (2010)	SELECT (2009)
Agent	Finasteride 5 mg	Dutasteride 0.5 mg	Selenium 200 µg ± vitamin E 400 IU
Men	18,882; PSA (prostate-specific antigen) ≤3	PSA 2.5–10, prior negative biopsy	35,533
Design	7 years; end-of-study biopsy	Biopsy at 2 and 4 years	Stopped early for futility
All cancers	↓24.8%	↓22.8%	No benefit
High grade	Gleason 7–10 ↑ (6.4% vs 5.1%)	Gleason 8–10 ↑ (0.9% vs 0.6%)	Vitamin E alone ↑ risk

The high-grade controversy

- Proposed artefacts: better PSA performance and a smaller gland make cancer easier to find.
- The FDA (Food and Drug Administration) did **not accept** these. In 2011 it refused a prevention licence and added a **high-grade warning** to 5ARI labels.
- PCPT at 18.4 years: non-significant 25% fewer prostate cancer deaths. Real-world data show no mortality effect.
- **Conclusion:** 5ARIs reduce detection of low-grade cancer, not death.

Vitamins and other agents

- **ATBC** trial: low-dose vitamin E cut prostate cancer 32% in smokers (secondary end point). SELECT then failed.
- **VITAL** (vitamin D): no prostate effect. High-dose zinc, calcium and antioxidants may harm.
- **Aspirin/NSAIDs** (non-steroidal anti-inflammatory drugs): small effect. **Statins:** observational benefit only. **Metformin:** none.
- Diet: whole foods; avoid obesity.

Current position

- No agent is approved for prevention. 5ARIs may be discussed with men taking them for BPH (benign prostatic hyperplasia).

Diagram to draw: bar chart: all cancers ↓ vs high-grade ↑ in PCPT and REDUCE.

> **Key points to remember:**
> - PCPT ↓24.8%; REDUCE ↓22.8%; both ↑ high grade.
> - FDA warning; no approved preventive agent.
> - SELECT negative; vitamin E alone harmful.

IV-3b. Prepucioplasty.

Type: SAQ (10 marks)

Read as: preputioplasty, a foreskin-preserving operation for phimosis.

Definition

- An operation that widens the tight preputial ring but keeps the foreskin. It is an alternative to circumcision.

Background

- **Physiological phimosis** at birth is due to glans–preputial adhesions. Smegma and erections separate them. About 90% retract by 3 years; fewer than 1% are phimotic at 17 years.
- **Secondary phimosis** follows scarring (forced retraction, balanitis) or BXO (balanitis xerotica obliterans).

Indications

- Persistent symptomatic phimosis: recurrent balanoposthitis, UTI (urinary tract infection), paraphimosis, painful erection.
- Failed **topical steroid** (betamethasone 0.05% twice daily for about 4 weeks). About 1 in 6 fail.
- Wish to keep the foreskin (cultural reasons, cosmesis).

Contraindications

- **BXO** with scarring: circumcision is preferred. Triradiate incision with triamcinolone has been reported in selected boys.
- Physiological non-retractility needs only reassurance. Treat active infection first.

Techniques

Technique	Method
Dorsal slit, transverse closure	Longitudinal incision through the ring, closed transversely (Heineke–Mikulicz)
Triple incision	Three longitudinal cuts over the ring, each closed transversely
Y–V plasty	V-flap of outer skin advanced into a Y-cut

- General anaesthesia or penile block; confirm full retraction; absorbable sutures.

Postoperative care

- **Daily retraction from the first week**, continued for weeks, prevents re-stenosis. A topical steroid may help.

Outcomes and complications

- Most achieve a retractile foreskin. It is quicker and less painful than circumcision.
- Complications: re-phimosis (needs circumcision), bleeding, infection, poor cosmesis, paraphimosis.
- **Also know:** preputial reconstruction in hypospadias repair is also called preputioplasty; dehiscence is its risk.

Diagram to draw: dorsal incision through the ring, closed transversely.

Key points to remember:

- Steroid first; about 1 in 6 fail.
- Cut lengthwise, close crosswise; daily retraction after.
- BXO: circumcision preferred.

IV-3c. Optical coherence tomography in carcinoma bladder.

Type: SAQ (10 marks)

Introduction

- WLC (white-light cystoscopy) misses flat lesions and cannot show depth.
- **OCT (optical coherence tomography)** is an "optical biopsy": real-time cross-sectional imaging of the bladder wall. It works like ultrasound but uses light.

Principle

- **Near-infrared light** (about 1300 nm) is shone on tissue. Back-scattered light is measured by **low-coherence interferometry** against a reference beam.
- The delay of reflected light gives depth. Scanning across builds a cross-sectional image.
- **Resolution about 10–20 µm** (near histology). **Depth about 2–3 mm**: urothelium, lamina propria and superficial muscle.
- A probe passes through the cystoscope channel and touches the mucosa.

Appearance

Normal	Tumour
Thin, dark urothelium	Thickened urothelium
Bright lamina propria; clear boundary	Layers lost
Muscle visible below	Broken boundary = invasion (T1)

Uses

- Tell tumour from inflammation at suspicious areas; fewer unnecessary biopsies.
- **Staging Ta vs T1** (lamina propria invasion); helps in diverticula and thin walls.
- Check margins and completeness of TURBT (transurethral resection of bladder tumour).

Accuracy and limitations

- High sensitivity; moderate specificity. Inflammation, scar, recent TURBT or BCG (bacille Calmette-Guérin) give false positives. Adding fluorescence cystoscopy improves specificity.
- Small field: needs WLC or PDD (photodynamic diagnosis) to guide it.
- Cannot judge deep muscle invasion; grading unreliable; cost.
- Still investigational; not in routine guideline use.

Other enhanced cystoscopy

- **PDD (blue light, hexaminolevulinate):** tumour porphyrins fluoresce red; finds about one-third more CIS (carcinoma in situ).
- **NBI (narrow-band imaging):** 415/540 nm light shows vessels; many false positives.
- **Confocal laser endomicroscopy:** fluorescein; cell-level surface image.
- **Also know:** Raman spectroscopy.

Diagram to draw: OCT image of normal layers vs tumour with lost layers.

> **Key points to remember:**
> - Near-infrared interferometry; 10–20 µm resolution; 2–3 mm depth.
> - Role: Ta vs T1; cannot assess deep muscle.
> - High sensitivity, moderate specificity; investigational.

IV-3d. Re-look TUR bladder tumour.

Type: SAQ (10 marks)

Definition

- Second (restaging) TURBT (transurethral resection of bladder tumour) of the primary site, including detrusor, a few weeks after the first. It is a quality indicator of NMIBC (non-muscle-invasive bladder cancer) care.

Rationale

- **Incomplete resection:** residual tumour at the site in ≥40%; tumour at first check cystoscopy in up to 45%.
- **Understaging:** in HG (high-grade) T1, re-resection shows worse disease or muscle invasion in 25–30%. This rises to about half if the first specimen had **no muscle**.

Indications (AUA [American Urological Association] / EAU [European Association of Urology])

- **Recommended:**
  - Incomplete first resection.
  - **All T1.**
  - No muscle in specimen (except low-grade Ta and primary CIS [carcinoma in situ]).
- **Consider:** large or multifocal HG Ta (residual up to 50%, upstaging up to 15%).

Not needed

- Low-grade Ta, primary CIS, clear muscle invasion, or planned cystectomy.

Timing and technique

- **2–6 weeks** after the first TURBT (AUA: within 6 weeks), before BCG (bacille Calmette-Guérin).
- Resect the scar and base deep into detrusor; send the base separately. Use blue light or NBI (narrow-band imaging) if available.

Benefits

- Corrects stage and **changes treatment in about one-third**.
- **Divrik RCT (randomised controlled trial, 2010):** re-TUR in T1 improved recurrence-free and progression-free survival.
- Better BCG response with minimal residual tumour.

Result-based plan

- **Residual T1** means progression up to 80%: consider early cystectomy.
- T0 or Ta: BCG induction + maintenance. T2: treat as muscle-invasive.

Recent advances

- **En bloc resection** gives muscle more often and may reduce re-TUR. It is not yet a proven substitute.

Diagram to draw: flowchart: T1 or no muscle → re-TUR 2–6 weeks → T2: cystectomy pathway / residual T1: consider cystectomy / T0–Ta: BCG.

Key points to remember:

- Re-TUR: all T1, incomplete resection, no muscle; consider HG Ta.
- 2–6 weeks; residual ≥40%; upstaging 25–30% (half if no muscle).
- Residual T1 → progression up to 80%.

IV-3e. Male menopause.

Type: SAQ (10 marks)

Read as: late-onset hypogonadism (testosterone deficiency of ageing men, "andropause").

Definition

- Symptoms plus **consistently low serum testosterone** in an ageing man.
- "Menopause" is a misnomer: the fall is gradual and partial. Preferred terms: PADAM (partial androgen deficiency of the ageing male), LOH (late-onset hypogonadism).

Pathophysiology

- Testosterone in the 70s is about one-third lower than in young men.
- **Testis:** fewer Leydig cells. **Centre:** blunted feedback, irregular GnRH (gonadotrophin-releasing hormone) pulses.
- **SHBG (sex hormone-binding globulin) rises**, so free testosterone falls more.
- Worse with obesity, diabetes and opioids.

Clinical features

- **Sexual (most specific):** low libido, ED (erectile dysfunction), fewer morning erections.
- Fatigue, low muscle mass, osteoporosis, anaemia, low mood.
- **ADAM (Androgen Deficiency in the Aging Male)** questionnaire: sensitive, not specific.

Diagnosis

- **Morning (7–11 am) total testosterone, twice.** AUA (American Urological Association): **<300 ng/dL**. EAU (European Association of Urology): about **12 nmol/L**; calculated free testosterone if borderline.
- **LH (luteinising hormone):** high in primary, low or normal in secondary hypogonadism. Check **prolactin** if LH is low.
- Baseline: PSA (prostate-specific antigen), DRE (digital rectal examination), haematocrit.

Treatment

- Lifestyle first: weight loss, exercise.
- **TRT (testosterone replacement therapy)** for symptoms + confirmed low level: gels, injections (enanthate every 2–3 weeks; undecanoate every 10–14 weeks), oral undecanoate. Stop if no benefit at 3–6 months.
- **Contraindications:**
  - Prostate or breast cancer.
  - **Wish for fertility** (TRT stops sperm production; use hCG [human chorionic gonadotrophin] or clomiphene).
  - Haematocrit >50%.
  - Untreated severe sleep apnoea; recent MI (myocardial infarction) or stroke.

Monitoring

- At 3–6 months, then yearly: testosterone, **haematocrit** (stop if >54%), PSA.

Safety evidence

- **TRAVERSE (2023):** 5,246 men aged 45–80 at high cardiovascular risk. TRT was **non-inferior for major cardiac events**; more atrial fibrillation and pulmonary embolism.

Diagram to draw: hypothalamic–pituitary–testis axis with ageing defects.

Key points to remember:

- Symptoms + low morning testosterone twice (AUA <300 ng/dL; EAU 12 nmol/L).
- LH separates primary from secondary; prolactin if LH low.
- TRT harms fertility; watch haematocrit; TRAVERSE: cardiac non-inferiority.

IV-3f. Acute urinary retention in females.

Type: SAQ (10 marks)

Introduction

- AUR (acute urinary retention) is sudden inability to void. It is uncommon in women and is often non-obstructive, so search for the cause.

Causes

Group	Examples
Obstructive	Large cystocele or prolapse; tight sling after incontinence surgery; urethral diverticulum, stricture; pelvic mass, **retroverted gravid uterus**, haematocolpos; cancer
Functional	**Fowler syndrome**; dysfunctional voiding; primary bladder neck obstruction (rare)
Neurogenic	MS (multiple sclerosis; may present first with retention), cauda equina, transverse myelitis, diabetes
Inflammatory	Cystitis, vulvovaginitis, **genital herpes**, Bartholin abscess
Drugs	Anticholinergics, opioids, tricyclics, calcium-channel blockers
Postoperative/postpartum; psychogenic	Epidural, pain; diagnosis of exclusion

Fowler syndrome

- Young women (20–35) with painless retention and bladder capacity over 1 litre.
- **Urethral sphincter EMG (electromyography):** complex repetitive discharges, so the sphincter fails to relax. Urethral closure pressure >100 cm H_2O.

Evaluation

- **History:** drugs, surgery, delivery, neurological and bowel symptoms, genital lesions.
- **Examination:** pelvic (prolapse, mass); neurological (perineal sensation, anal tone).
- **Tests:** drained volume; urine culture; creatinine; pregnancy test; ultrasound.
- MRI (magnetic resonance imaging) spine if neurological signs; cystoscopy.
- **Later:** video-urodynamics (female obstruction, e.g. Qmax ≤12 mL/s with Pdet ≥20 cm H_2O); sphincter EMG.

Management

- **Immediate:** urethral catheter; suprapubic if urethral access fails.
- **Treat the cause:** stop drugs; treat infection or herpes; pessary or prolapse repair; reposition the uterus; sling division; diverticulectomy.
- **TWOC (trial without catheter)** after correction.
- **Persistent retention:**
  - **CISC (clean intermittent self-catheterisation)** rather than an indwelling catheter.
  - **SNM (sacral neuromodulation):** about 70% success in Fowler syndrome.
  - Bethanechol not recommended. Bladder neck incision only with caution (incontinence).

Diagram to draw: flowchart: AUR → catheterise → find cause → treat → TWOC → CISC or SNM.

Key points to remember:

- Look for prolapse, sling obstruction, pelvic mass, neurology and drugs.
- Fowler: abnormal sphincter EMG; SNM about 70%.
- CISC for persistent retention.

Rapid Revision Sheet – Paper IV

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

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- Single instillation within 6 h; never BCG immediately (SWOG S0337 gemcitabine).
- Only maintenance BCG reduces progression; SWOG 3-year schedule.
- BCG-unresponsive: cystectomy; pembrolizumab, nadofaragene, N-803.

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- Nd:YAG for PeIN; eyewear for everyone.

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- PCPT ↓24.8%; REDUCE ↓22.8%; both ↑ high grade.
- FDA warning; no approved preventive agent.
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- Steroid first; about 1 in 6 fail.
- Cut lengthwise, close crosswise; daily retraction after.
- BXO: circumcision preferred.

IV-3c. Optical coherence tomography in carcinoma bladder. (SAQ)

- Near-infrared interferometry; 10–20 µm resolution; 2–3 mm depth.
- Role: Ta vs T1; cannot assess deep muscle.
- High sensitivity, moderate specificity; investigational.

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